Math 226 Final Exam
Fa25
Wed Dec 10

Firstname Lastname: ______________________________

USC Email: ______________________ UscID: ______________________

Instructions.

- This examination consists of 19 pages not including this cover page.
- Write your initials in the designated spot at the top–right of each page that contains work which you would like to have graded.
- This examination consists of 9 questions for a total of 100 points. You have 120 minutes to complete this examination.
- Do not use books, calculators, computers, tablets, or phones.
- You may use a single 8.5 in by 11 in page of notes, handwritten on both sides.
- Write legibly in the boxed area only. Cross out any work that you do not wish to have scored.
- Show all of your work and cite theorems you use. Unsupported answers may not earn credit.
- If you run out of space: there is an extra page after each problem, and one extra page at the end. Please indicate on the page containing the original question when you are continuing your work on another page.
- All work you submit should represent your own thoughts and ideas. If the graders suspect otherwise: you can expect your instructor to file a report with USCs Office of Academic Integrity (OAI).

Circle your instructor: Prof Geske Prof Haine Prof Reyes Souto Prof Ziane

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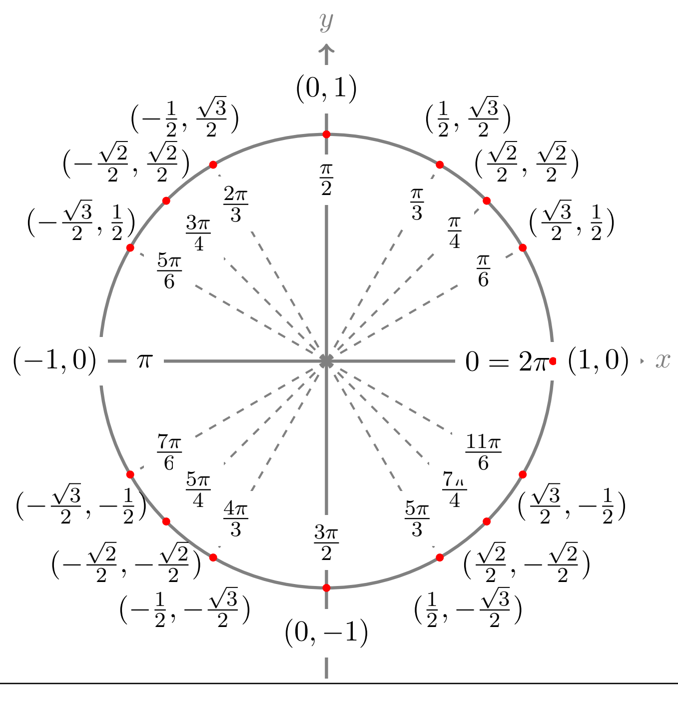
Question:	1	2	3	4	5	6	7	8	9	Total
Points:	10	12	10	10	10	10	12	12	14	100



$\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$

$\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$

$\sin 2\theta = 2\sin\theta\cos\theta$

1. (10 points) Consider the parametric curve $\mathbf{r}(t) = \langle \sin(t), \cos(3t), 4t^2 \rangle$ with domain $(-\infty, \infty)$.

   (a) Give a parametrization of the line L tangent to this curve at the point $P(1, 0, \pi^2)$.

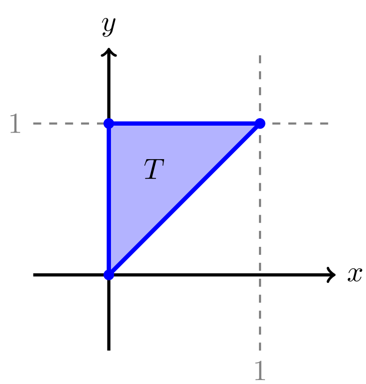
   (b) Write down the Cartesian equation (aka scalar equation) for the plane passing through P (from part a) and orthogonal to L (also from part a).

You may continue your work for Q1 on this page.

2. (12 points) Let $f(x, y) = 2x - xy - xy^2 + y^3$.

   (a) Find, and classify, all critical points of $f(x, y)$.

   (b) Find the absolute maximum and minimum values of f on the closed triangular region T depicted below.



You may continue your work for Q2 on this page.

3. (10 points) Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = x^2y + 1$ subject to $2x^2 + 4y^2 = 3$.

You may continue your work for Q3 on this page.

4. (10 points) Consider the solid E bounded by the surface $y^2 = x(2-z)$ and the plane $x+z=2$, and contained in the first octant, $x, y, z \geq 0$. Assume x, y, z are measured in cm. The density of the solid E at point (x, y, z) is given by $\delta(x, y, z) = 2y\ g/\text{cm}^3$. Calculate the total mass of E.

   Hint: an integral in order $dy d\star d\star$ may be most appropriate.

You may continue your work for Q4 on this page.

5. (10 points) Consider the region R defined by the inequalities:

$$x^2 + y^2 + z^2 \leq 9 \quad \text{and} \quad -\frac{1}{\sqrt{3}}\sqrt{x^2+y^2} \leq z \leq 0$$

Compute the value of the triple integral:

$$\iiint\limits_R \frac{\arcsin\left(\frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}\right)}{\sqrt{x^2+y^2}}\, dV$$

Hint: recall that, if $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$, then $\arcsin(\sin t) = t$. Recall also that $\sin(t) = \sin(\pi - t)$.

Hint: this integral may look scary as presented, but in spherical coordinates looks far less scary.

You may continue your work for Q5 on this page.

6. (10 points) Consider the vector force field, with domain the half–plane $x + y > 0$, given by:

$$\mathbf{F}(x, y) = \left[ye^x + \frac{y}{x+y} \right] \mathbf{i} + \left[2 + e^x + \frac{y}{x+y} + A\ln(x+y) \right] \mathbf{j}$$

(a) There is a value of constant A so that $\mathbf{F}$ is conservative on its domain. Find that value.

(b) Using the value of A you found in part (a), find a potential f for $\mathbf{F}$.

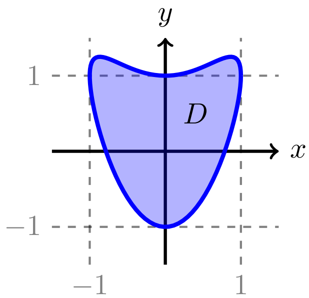
(c) Using the value of A you found in part (a), evaluate the work done by the conservative force field $\mathbf{F}$ on a particle moving along the path:

$\mathbf{r}(t) = \left\langle \cos^4 t, \frac{2t}{\pi} \right\rangle$ with $0 \le t \le \pi$

You may continue your work for Q6 on this page.

7. (12 points) The path:

$\mathbf{r}(t) = \left\langle \cos t, \cos^2 t + \sin t \right\rangle$ with $0 \leq t \leq 2\pi$ encloses the following region D.



(a) Use Green's Theorem, by computing an appropriate line integral, to find the area of the region D.

Hint: there are trig identities on the cover page.

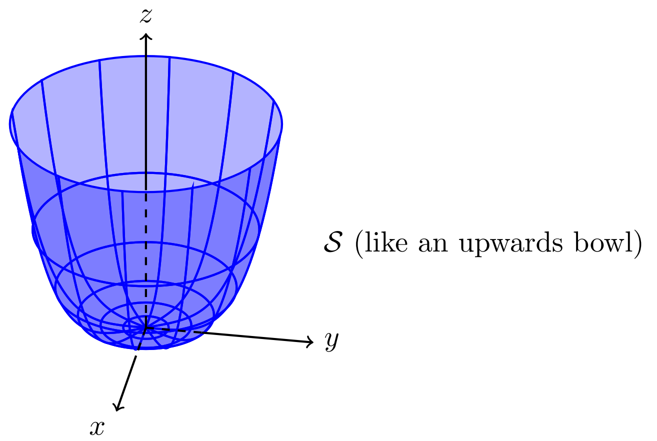
(b) Find:

$$\oint_{\mathbf{r}} \left[e^{x^2} + y^2 + 2y \right] \, dx + \left[2xy + \sin\left(y^2\right) \right] \, dy$$

Note: if you would like to use the answer to part (a) for part (b), but were unable to solve part (a), then use symbol A to represent the answer to part (a).

You may continue your work for Q7 on this page.

8. (12 points) Consider the surface $\mathcal{S}$ equal to the portion of $z = (x^2+y^2)^2$ with $z \leq 1$. Orient $\mathcal{S}$ with upwards normals.



Consider also the vector field:

$\mathbf{F}(x, y, z) = \langle z, x, \sin^2(z-1) \rangle$

Confirm Stokes's Theorem for this $\mathcal{S}$ and $\mathbf{F}$ by computing a relevant surface integral and a relevant line integral, and confirming equality. This is divided into parts below.

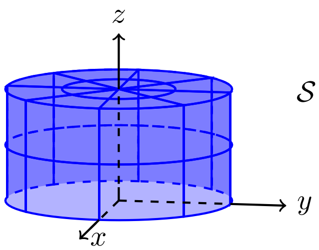
(a) For this part, directly compute the **surface** integral relevant to Stokes's Theorem. Note: you must use **surface** integral, not line integral, calculations here.

Hint: one approach is to use polar coordinates as parameters for $\mathcal{S}$. Be careful, you are not computing the surface integral of $\mathbf{F}$ here, rather, you are computing the surface integral of ......

(b) For this part, directly compute the **line** integral relevant to Stokes's Theorem. Confirm that your answers to the two parts are equal. Note: you must use **line** integral, not surface integral calculations here.

You may continue your work for Q8 on this page.

9. (14 points) Let $\mathcal{S}$ consist of the cylinder defined by $x^2 + y^2 = 1$ with $0 \leq z \leq 1$ along with its **top** $x^2 + y^2 \leq 1$ with $z = 1$. Orient $\mathcal{S}$ with outward normals.



$\mathcal{S}$ (top included, bottom not included)

(a) The surface $\mathcal{S}$ is not closed. Describe the simplest surface $\mathcal{B}$ that you would have to join to $\mathcal{S}$ to obtain a closed surface. To be compatible with outward normals for the closed surface: should the normals to $\mathcal{B}$ point up or down?

(b) Find $\iint_{\mathcal{B}} \mathbf{F} \cdot d\mathbf{S}$ where $\mathcal{B}$ is from part (a) and:

$$\mathbf{F} = \left(x + ye^{z^3}\right)\mathbf{i} + \left(z^2\cos(x^2) - y\right)\mathbf{j} + \left(1 + z\right)\mathbf{k}$$

(c) Using a combination of the divergence theorem and your earlier work, find:

$$\iint_{\mathcal{S}} \mathbf{F} \cdot d\mathbf{S}$$

where $\mathbf{F}$ is from part (b) and $\mathcal{S}$ is from the top of the page.

You may continue your work for Q9 on this page.

If you would like work on this page scored, then clearly indicate to which question the work belongs and indicate on the page containing the original question that there is work on this page to score.